

EDUCATION

- Northerwestern University Evanston, IL, USA September 2023 – Present
Degree: Ph.D.; Chemical and Biological Engineering
- University of Victoria Victoria, BC, CAN January 2020 – April 2022
GPA: 8.0 / 9.0 (4.0 / 4.0 eqv.) **Degree:** M.A.Sc.; Civil Engineering Major
- Grand Valley State University Allendale, MI, USA August 2016 – April 2019
GPA: 3.6 / 4.0 **Degree:** B.S. (honors); Chem. major, Bio. minor, Green Chem. certificate
- Grand Rapids Community College Grand Rapids, MI, USA August 2015 – August 2016
GPA: 3.8 / 4.0 **Degree:** A.S. (honors)
- East Grand Rapids High School East Grand Rapids, MI, USA September 2011 – May 2015
GPA: 3.7 / 4.0; 4.22 weighted **Degree:** Diploma (honors)

RESEARCH

- Northwestern University, Keith Tyo's Lab October 2023 – Present
- **Graduate researcher** of microbial ecology.
- Argonne National Lab, Christopher Henry's Group June 2021 – Present
- **Assistant Computational Biologist** (metabolic modeling, bioinformatics, and ML)
 - Contributed to the [KBase](#) and [PMI](#) SFAs & various collaborator projects.
- University of Victoria, Heather Buckley's Group January 2020 – April 2022
- **Software engineer** for models of bacterial biochemistry and desalination geochemistry
- Mount Sinai School of Medicine, Jonathan Karr's Lab December 2020 – April 2022
- **Data scientist** and bioinformatician for [Biosimulators.org](#) and [Biosimulations.org](#)
- Lawrence Berkeley National Lab, Reactive Transport Group June 2020 – October 2020
- **Software engineer** of geochemical reactive transport during desalination
- North Carolina State University, Lucian Lucia's Group May 2019 – December 2019
- **Metabolomics spectroscopist** of *Cannabis* – w/ Shaw Univ. and a startup GenoVerde Biosciences
- Washington State University, Michael Wolcott's Group May 2018 – August 2018
- **NSF-funded REU** researcher of photocatalyzed oxygen-scavenging cellulosic materials
- Grand Valley State University, Dalila Kovacs' Group October 2016 – June 2019
- **Analytical chemist** of willow metabolomics to identify applications as an alternative feedstock.

EXPERIENCE

Webmaster

Assistant Webmaster, ACS Division of Organic Chemistry June 2020 – Present

- Maintainer of www.organicdivision.org and www.organicchemistrydata.org
- 2022 ACS ChemLuminary Recipient

Webmaster, ACS Division of Geochemistry

July 2022 – Present

- Maintainer of <https://www.acsgeoc.org/>

Webmaster, Jewish Studies programs

June 2023 – May 2024

- Maintainer of www.studyjudaism.net ; www.convertingtojudaism.net ; www.moodtek.com

Teaching

Instructor

- University of Victoria May 2021 – August 2021
 - CIVE 210 Sustainable Design

Teacher's Assistant

- Northwestern University March 2024 – Present
 - CHEM ENG 210 Analysis of Chemical Process Systems
- University of Victoria January 2020 – April 2021
 - ECE 340 Applied Photonics and Electromagnetism
 - CIVE 345 Fluid Mechanics
 - CIVE 210 Sustainable Design
 - CHEM 150 and 102 laboratories
- Grand Valley State University August 2018 – December 2018
 - CHM 109 discussion section

Tutor

- Shaw University September 2019 – November 2019
 - Organic Chemistry
- Grand Valley State University August 2017 – April 2019
 - College Reading & Learning Association **certified tutor** of general and Organic Chemistries
- High School Tutor February 2017 – November 2020
 - Chemistry, Physics, Biology, Mathematics, and Writing

Science Writer

Freelance Science Writer

December 2019 – March 2021

- *Canadian Vegan* magazine (sampled here www.andrewfreiburger.com/reviews)

Other

Fresh Thyme Farmer's Market Dairy Associate

May – August 2016

- Independently operated the dairy department

Sports and Recreation

- East Grand Rapids Recreation Department
 - Elementary school “Safety Town” Program June 2014 – August 2014
 - Soccer and Basketball scorekeeper and referee August 2011 – August 2016
- Southern Little League Baseball Umpire June 2010 – August 2010

SKILLS

Computer

Python (data science, optimization, ML); HTML & CSS; Git & GitHub; linux; command prompt; Java; C++; PHREEQ(C/Py); Juliat; Microsoft office; WordPress

Instruments

¹H NMR, ¹³C NMR, 2D-NMR, GC-MS, GC-FID, HPLC, UV-Vis, FTIR, ICPMS, Bomb Calorimeter, Lyophilizer, Muffle Furnace, Biological Safety Cabinet

Experiments

Matrix casting, Soxhlet Extraction, Steam Distillation, Ball milling, Dual Asymmetric Centrifugation, Plant Cloning, Risk Group II Bacterial Culturing

PRESENTATIONS

Oral (20-minute) Presentations

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|---|-----------|---------------|
| • ACS Spring National Conference | (GA) | Spring 2026 |
| • AIChE National Conference | (MA) | Fall 2025 |
| • ACS Fall National Conference | (DC) | Fall 2025 |
| • Genome-Resolved Open Watersheds workshop | (CO) | March 2024 |
| • ACS Spring National Conference | (IN) | March 2023 |
| • ACS Spring National Conference | (virtual) | March 2022 |
| • PacifiChem | (virtual) | December 2021 |
| • (2 talks) ACS Spring National Conference | (virtual) | April 2021 |
| • Central Canadian Association on Water Quality (56 th) | (virtual) | March 2021 |
| • ACS Green Chemistry and Engineering (23 rd) | (VA) | June 2019 |
| • GVSU Student Scholar’s Day | (MI) | April 2019 |

Lightning Talks (3-minute)

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|---|-----------|-------------|
| • LBNL Molecular Foundry User Meeting | (virtual) | August 2020 |
| • ACS Fall National Conference (260 th) | (virtual) | August 2020 |
| • ACS Green Chemistry and Engineering (24 th) | (virtual) | June 2020 |

Poster Presentations

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|---|-----------|---------------|
| • GoldSchmidt | (IL) | August 2024 |
| • ACS Fall National Conference | (CO) | August 2024 |
| • DOE GSP PI’s Meeting | (MD) | April 2024 |
| • ACS Fall National Conference | (CA) | August 2023 |
| • ASM National Conference | (TX) | June 2023 |
| • ACS Spring National Conference | (IN) | March 2023 |
| • AIChE National Conference | (AZ) | November 2022 |
| • ACS Fall National Conference | (IL) | August 2022 |
| • ACS Spring National Conference | (virtual) | March 2022 |
| • Central Canadian Association on Water Quality (55 th) | (virtual) | July 2020 |

• Michigan Forest Bioeconomy conference (2 nd)	(MI)	February 2019
• Center for Undergraduate Research REU symposium	(VA)	October 2018
• Washington State University Summer research symposium	(WA)	August 2018
• ACS Green Chemistry and Engineering (22 nd)	(OR)	June 2018
• GVSU Student Scholar's Day (22 nd)	(MI)	April 2018

* A complete list, with full citations, of my presentations are provided on my website
(www.andrewfreiburger.com/CV/conferences.html)

ACCOLADES

NSF Innovation Corps (i-corps) Bay Area graduate	June 2020
ACS Division of Organic Chemistry Undergraduate Award	April 2019
Grand Valley State University Honor's college	Fall 2016 – April 2019
Grand Rapids Community College Honor's college	Fall 2015 – April 2019
Grand Valley State University Sustainability Award	April 2019
(7x) Dean's List	Fall 2015 – Present
President's List	1 Semester
6 th in National "We the People", Center for Civic Ed., competition	Winter 2014
Michigan Youth Leadership school representative	June 2013
1 st Chair Trumpet in Concert Band	2012 – 2013
Summa Cum Laude	Fall 2011 – Spring 2015

SCHOLARSHIPS

ACS GEOC Student Travel Award	April 2021
(3x) University of Victoria Graduate Award	W., Spr., Sum. 2020
Purdue University Graduate School Expo travel scholarship	September 2019
Academic and Professional Enrichment Fund award	June 2018, 2019
Great Finish Grand Scholarship	Fall 2018

OUTREACH

SLAC Science Accelerating Girls' Engagement in STEM (SAGE-S) camp	August 2020
GVSU Networks of Support K-12 Tutoring	June 2020 – August 2020
LBNL "Homework Help Volunteers" K-12 Education	April 2020 – May 2020
British Columbia Virtual Science Fair judge and Mentor	June 2020
Oakland Science and Engineering Fair judge and Mentor	May 2020
Article reviewer for the journal of Wood and Fiber Science	September 2019
Organic chemistry laboratory teaching assistant at Shaw University	September 2019
Science Olympiad competition volunteer	March 2019
"Fall in love with STEM" Valentine's day elementary student volunteer	February 2019
ACS "Chemistry at the Mall" demonstrations	October 2017, 2018
Guest lecturer for a 100-level chemistry class	October 2017

PERSONAL

Chabad community volunteer	October 2023 – Present
Volunteer chemist for the OpenAir Collective DACC initiative	July 2020 – August 2021
Chinese Martial Artist	August 2015 – January 2020
Photography model	August 2017 – August 2020
Band leader and gig musician	January 2018 – July 2020
3 rd degree Knight of Columbus	October 2018
Groundswell and Green Wagon Organic Farm volunteer	Summers 2015, 2016

“Hearts of Gold” charity student representative and public communicator	October 2012 – April 2015
Michigan Youth Leadership (MYLead) camp counselor	June 2014
100 hours as a “Safety Town” camp counselor	Summer 2013
Monthly Church volunteer	2009 – 2013

REFERENCES

Chris Henry, Computational Science Leader, Data Science and Learning, ANL
 – Research mentor (May 2021 – Present) chenry@anl.gov

Heather Buckley, Associate Professor, Department of Civil Engineering, UVic
 – Research mentor (December 2019 – April 2022) hbuckley@uvic.ca

Lucian Lucia, Professor, Department of Forest Biomaterials, NCSU
 – Research mentor (May 2019 – December 2019) lalucia@ncsu.edu

James Krikke, Professor, Department of Chemistry, GVSU
 – Research mentor (December 2016 – June 2019) krikkeji@gvsu.edu

PUBLICATIONS

“A functional microbiome catalogue crowdsourced from North American rivers”. *Nature*. **2025**.
<https://doi.org/10.1038/s41586-024-08240-z>.

“Diet outperforms microbial transplant to drive microbiome recovery in mice”. *Nature*. **2025**.
<https://doi.org/10.1038/s41586-025-08937-9>.

“BioSimulators: a central registry of simulation engines and services for recommending specific tools”.
Nucleic Acids Research. **2022**. <https://doi.org/10.1093/nar/gkac331>.

“Mapping the soil microbiome functions shaping wetland methane emissions”. *mSystems*. **2026**.
<https://doi.org/10.1128/msystems.00680-25>.

“KBase: Open-source platform for collaborative biological data analysis and publication”. *Journal of Molecular Biology*. **2026**. <https://doi.org/10.1016/j.jmb.2026.169676>.

“Hemp is the 21st century Tobacco: A Review”. *ACS Journal of Agricultural Science and Technology*.
2021. <https://doi.org/10.1021/acsagscitech.1c00114>

*A complete list and full citations of my publications is provided on my website
<https://www.andrewfreiburger.com/CV/publications.html>*